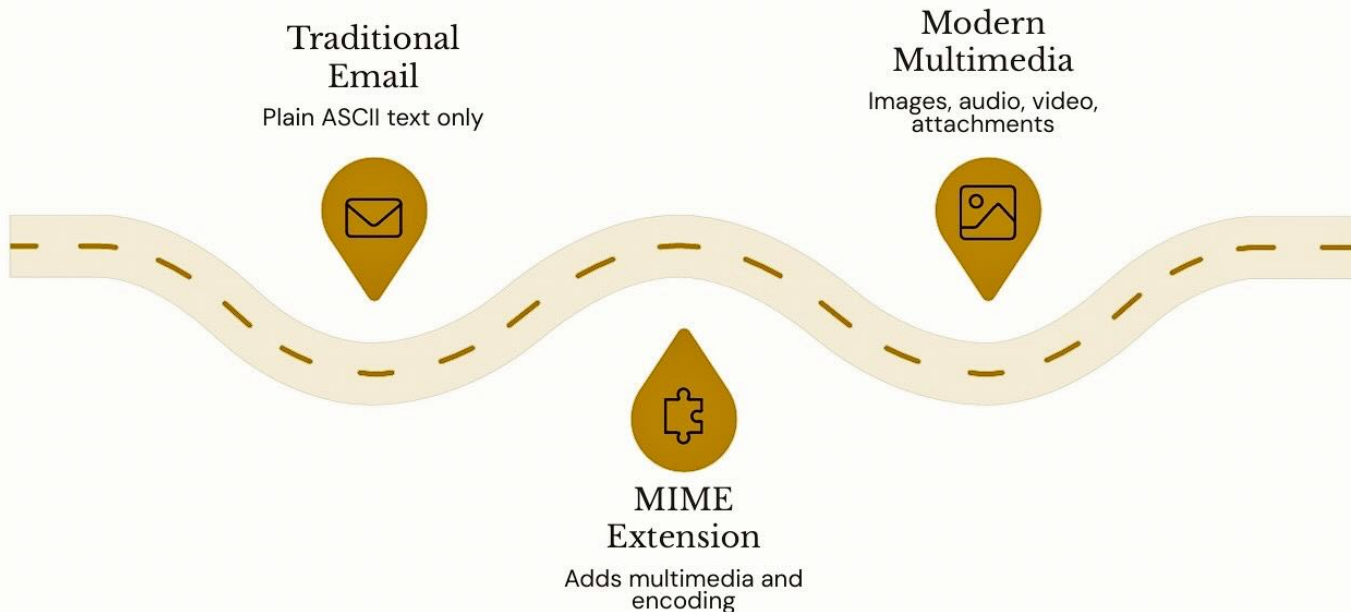


MIME — Multipurpose Internet Mail Extensions

Traditional email systems were designed for a simpler era, supporting only **plain ASCII text** with a strict 7-bit character limit. This meant no formatting, no attachments, no images, and no support for non-English character sets — a severe constraint that made email unsuitable for modern digital communication.

MIME was introduced as an **internet standard (RFC 2045–2047)** to solve this limitation. It functions as an extension layer that sits on top of standard email protocols, allowing a single message to contain multiple content types — text, images, audio, video, documents, and file attachments — seamlessly mixed together within one message envelope.

From ASCII to Multimedia: The Evolution of Email



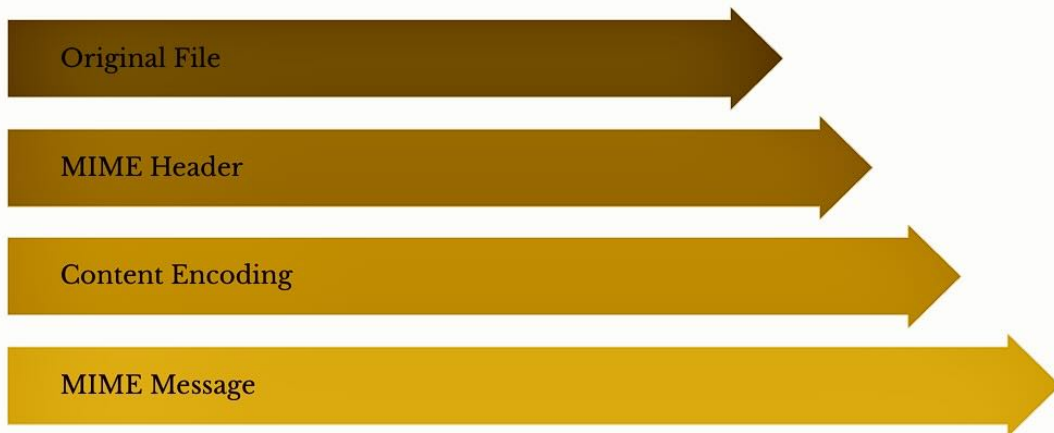
Supporting content types: text, images (JPEG/PNG), audio (MP3), video (MP4), documents (PDF/Word), and arbitrary file attachments.

Need and Components of MIME

MIME extends email headers with four key fields that allow receivers to correctly interpret message contents. Each header serves a specific purpose in the encoding and delivery pipeline.

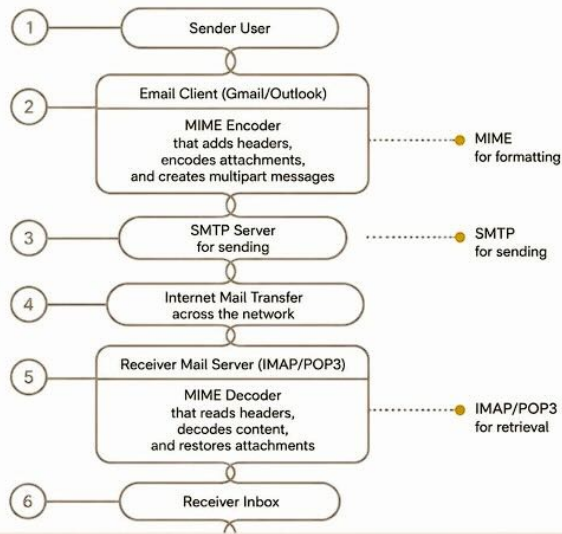
1	2
MIME-Version A simple header field (typically MIME-Version: 1.0) that flags the message as MIME-formatted, signaling the receiver to parse MIME-specific headers and multipart structures.	Content-Type Declares the media type of the payload using <i>type/subtype</i> notation. Examples: text/plain, text/html, image/jpeg, application/pdf, multipart/mixed.
3	4
Content-Transfer-Encoding Specifies how binary data is converted into a 7-bit-safe format for SMTP transport. Base64 encoding maps binary bytes into 64 printable ASCII characters, preserving file integrity across the network.	Content-Disposition Indicates whether content should be displayed <i>inline</i> within the email body or treated as an <i>attachment</i> with a filename — allowing email clients to render or save content appropriately.

MIME Message Construction Pipeline



MIME Email Architecture

The end-to-end journey of a MIME email traverses several protocol layers. Each layer serves a distinct role — encoding, transport, and retrieval — ensuring multimedia content moves reliably from sender to receiver.



SMTP — Sending

Transports the MIME-encoded message across mail servers using TCP port 25 or 587.

MIME — Formatting

Structures and encodes multimedia content into a format SMTP can safely carry.

IMAP/POP3 — Retrieval

Delivers the message to the receiver's client where MIME decodes and reconstructs it.

Complete MIME Mail Workflow

Consider a real scenario: a job applicant sends a resume. **Sender:** student@gmail.com → **Receiver:** hr@company.com. **Message:** "Please find my resume attached." **Attachment:** Resume.pdf

Step 1: Compose

The user writes the email in the client. The message body and PDF attachment are flagged for MIME processing before any data leaves the system.

Step 3: Attachment Encoding

The binary Resume.pdf is run through **Base64 encoding**, converting raw binary bytes into a safe stream of printable ASCII characters that SMTP can transport.

Step 2: MIME Divides Content

MIME creates a multipart/mixed message containing separate parts: the text message, optional HTML content, and the PDF attachment — each with its own Content-Type header.

Step 4: SMTP Transfer

The complete MIME message — headers, encoded body, and Base64 attachment — is transmitted via SMTP to the receiver's mail server for storage and retrieval.

Internal Structure & Advantages

Below is what a MIME message looks like internally. A **boundary string** separates each part, allowing the receiver's client to identify and process each content block independently — reconstructing the original files on the other end.

```
MIME-Version: 1.0
Content-Type: multipart/mixed;
boundary="message-section"

--message-section
Content-Type: text/plain

"Hello HR, Please find my resume."
--message-section
Content-Type: application/pdf
Content-Disposition: attachment;
filename="Resume.pdf"
Content-Transfer-Encoding: base64

JVBERi0KJQKJQKJQo...
--message-section--
```



Multimedia Support

Images, audio, video, and rich HTML formatting in a single message.



Global Languages

UTF-8 and international character sets built into the MIME standard.



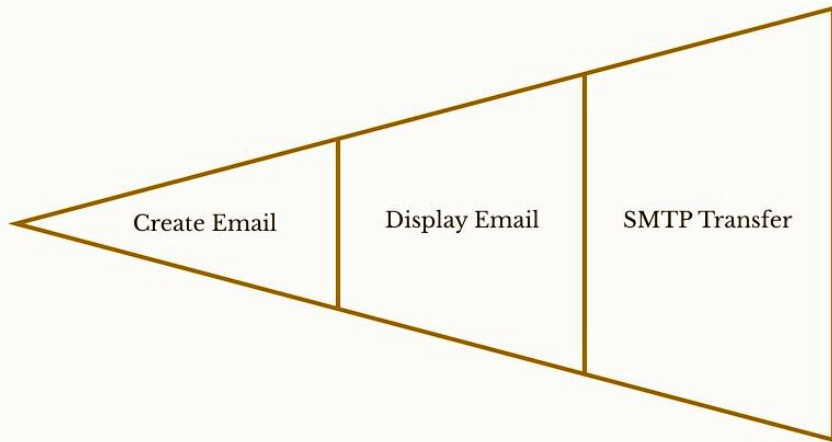
File Attachments

Any binary file — documents, archives, executables — via Base64 encoding.



Reliable Transfer

Encoding guarantees data integrity across SMTP, IMAP, and POP3.



☐ **MIME** is the packaging technology that transformed simple text emails into modern multimedia communication — enabling file attachments, formatted content, and multi-language support across every email protocol in use today.